



Diabetes Risk Prediction Using Machine Learning

CSP-571: Data Preparation and Analysis

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Diabetes Risk Prediction Using Machine Learning

1. Abstract

Diabetes is a chronic disease with substantial health and economic impacts worldwide. Early prediction of diabetes risk remains crucial for effective prevention and intervention. In this project, we developed a machine learning framework using the BRFSS 2015 dataset with 253,680 survey responses to classify individuals based on self-reported health indicators. We cleaned and prepared the data, addressed missing values and outliers, and applied the ROSE technique to balance the training set. We built and compared Logistic Regression, Decision Tree, Random Forest, and Neural Network models, evaluating them using Accuracy, Precision, Recall, and AUC. The Neural Network and Logistic Regression models achieved the strongest results, both with AUC around 0.82. Our work highlights the importance of data preparation, class balancing, and model comparison in building reliable healthcare prediction models. Future improvements will focus on tuning and feature engineering to enhance performance.

2. Overview

2.1 Problem Statement

Diabetes poses a critical public health challenge worldwide, imposing significant health and economic burdens. Recognizing the value of early identification, this project aims to create a predictive system using self-reported survey data to flag individuals at heightened risk for diabetes. Harnessing the power of machine learning enables us to move beyond traditional statistical models, offering more dynamic and scalable solutions.

2.2 Relevant Literature

Early predictions of diabetes has been widely studied using machine learning techniques, offering an improvement over traditional diagnostic methods. Several studies have applied models such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN) to predict diabetes risk based on health-related features.

One study highlighted that Logistic Regression achieved high accuracy (up to 97.2% with pipelining) when applied to diabetes prediction tasks, outperforming more complex clustering-based approaches. Surveys of machine learning applications for diabetes detection have consistently found that Random Forest and SVM models perform particularly well across diverse datasets like the PIMA Indian Diabetes dataset. Other studies have shown that techniques such as data cleaning, feature selection using Principal Component Analysis (PCA), and the use of association rule mining (e.g., the Apriori algorithm) can further improve predictive performance.

Overall, prior research has demonstrated that machine learning models are effective in predicting diabetes when careful data preparation and model selection are employed. These findings motivated our project to apply and compare multiple machine learning algorithms,

while also addressing challenges like class imbalance and feature engineering to enhance prediction reliability.

2.3 Proposed Methodology

Our methodological roadmap comprised the following stages:

- Data ingestion and preliminary cleaning.
- Exploratory data analysis to uncover structural properties.
- Feature extraction and engineering.
- Balancing classes using ROSE to mitigate bias.
- Model training and evaluation.
- Final model validation and interpretation.

3. Data Processing

3.1 Pipeline Details

We followed a structured data processing pipeline to prepare the BRFSS 2015 dataset for modeling. We started by ingesting the data and standardizing column names for consistency. We checked for missing values and verified that no placeholder codes like 999 were present, eliminating the need for imputation.

We flagged outliers in the dataset using the classic $1.5 \times \text{IQR}$ rule, which defines outliers as points falling below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$. The Interquartile Range (IQR) represents the middle 50% of the data and provides a robust method for detecting unusual values without being influenced by extreme observations. We removed extreme BMI values but preserved valid extreme responses in mental and physical health indicators to maintain the dataset's clinical relevance.

```
## Rows before: 253680 | Rows after: 243833 | Removed: 9847
```

Binary categorical variables were recast as factors, and continuous variables such as BMI, mental health days, and physical health days were standardized. Finally, we collapsed the diabetes outcome into a binary classification (diabetic vs. non-diabetic) and balanced the training set using the ROSE technique to ensure fair model training across classes.

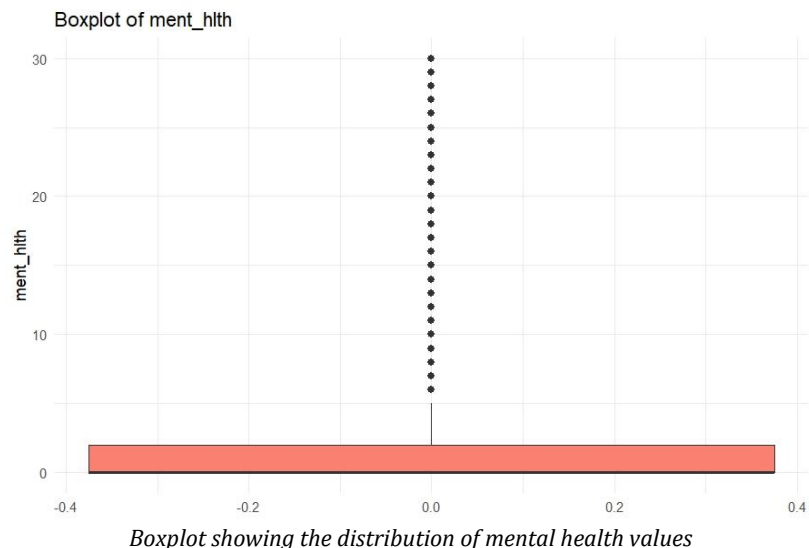
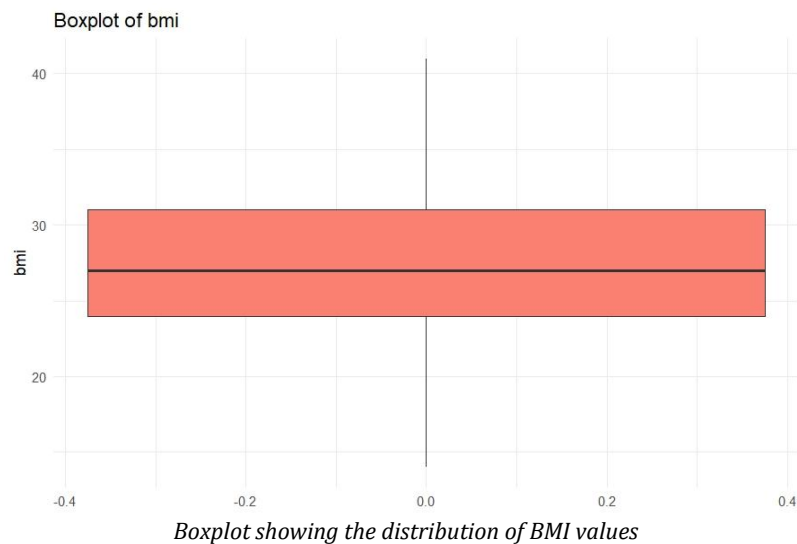
3.2 Data Issues

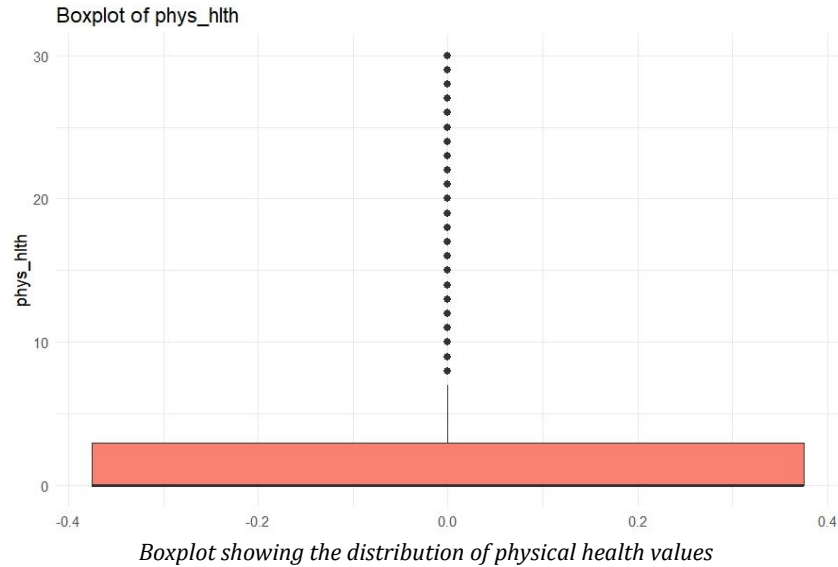
While the dataset was generally clean in terms of missing values, we encountered significant class imbalance. The original dataset was heavily imbalanced, with approximately 90% of individuals labeled as "No Diabetes," 2% as "Prediabetes," and 14% as "Diabetes." This imbalance posed a risk of models favoring the majority class and failing to identify diabetic cases accurately.

We also found that BMI values contained a small number of extreme outliers, and the distributions of mental and physical health days were heavily skewed. These patterns needed careful treatment to avoid distorting model behavior.

3.3 Assumptions/Adjustments

1. Based on IQR method, we observed that BMI values below approximately 13.5 or above 41.5 were rare and likely reflected measurement errors. In contrast, extreme values in the mental and physical health day variables, such as up to 30 days, appeared to be valid survey responses, often reflecting chronic health conditions. Therefore, we decided to remove outliers only from the BMI feature while preserving the full range of values for mental and physical health days. A set of boxplots was generated to visually inspect the distributions and verify the appropriateness of this decision.





2. Standardized continuous features and encoded categorical features for modeling.
3. We observed significant class imbalance in the original training set. To address this, we collapsed the outcome into a binary target (Diabetic vs. Non-Diabetic) and applied the ROSE technique to generate a balanced dataset with a roughly 50/50 class distribution, assuming that improving diabetic case detection was more critical than preserving the original proportions.

```
## Original training distribution:
print(table(train_data$diabetes_012))

##
## No Diabetes Prediabetes Diabetes
##      166012      3472    25584

## Balanced training counts:
table(balanced_train$diabetes_binary)

##
##      0      1
## 97792 97276
```

4. Data Analysis

4.1 Summary Statistics

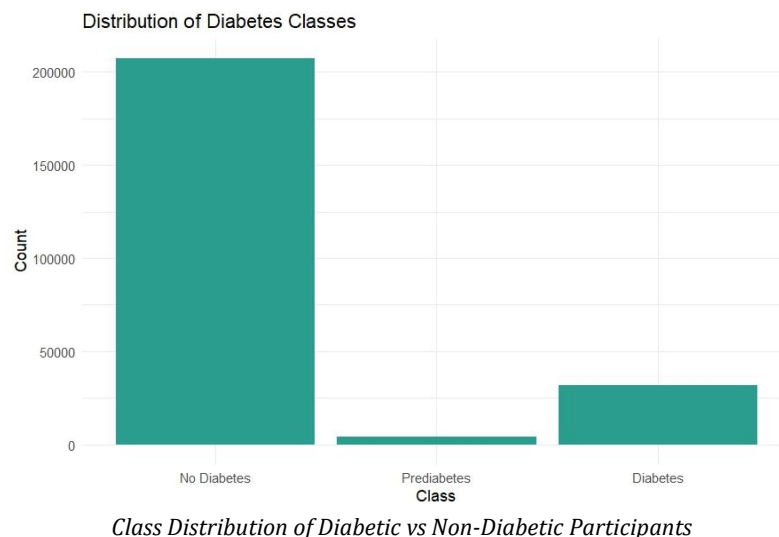
We analyzed the cleaned BRFSS 2015 dataset, which contained 243,833 records and 22 variables. The target variable, *diabetes_012*, had three classes: 0 (No Diabetes), 1 (Prediabetes), and 2 (Diabetes), with a strong class imbalance favoring non-diabetic individuals.

Among the continuous variables, BMI had a mean of 28.38, a median of 27.0, and ranged from 12.0 to 98.0. After outlier removal, the BMI distribution appeared approximately bell-shaped, centered between 25 and 30. Mental health days and physical health days were highly skewed toward zero, with medians of 0 but maximum values reaching 30, indicating a minority of individuals experiencing significant chronic health issues.

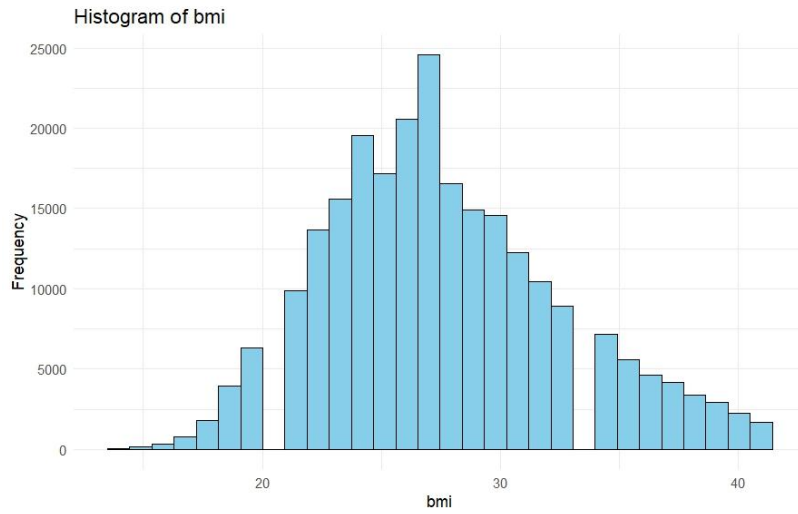
Demographic variables like age were uniformly distributed among adults, with a mean value of 8.03 (on a 1–13 scale corresponding to age groups), while education and income were right-skewed, with mean values of 5.05 and 6.05 respectively.

Binary health indicator variables such as high blood pressure, high cholesterol, smoking status, physical activity, and healthcare access were mostly skewed toward positive health behaviors, but with considerable variation across individuals. Boxplots and histograms were generated during exploratory analysis to visualize these distributions and inform preprocessing decisions like outlier treatment and feature scaling.

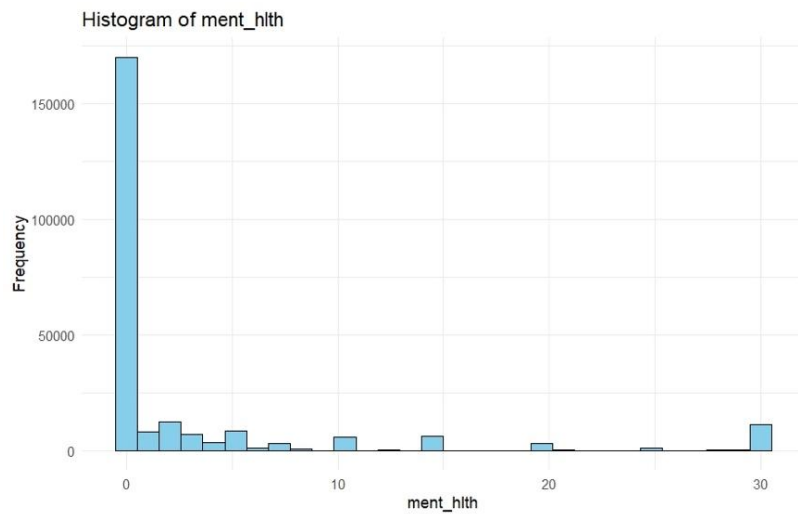
4.2 Visualization



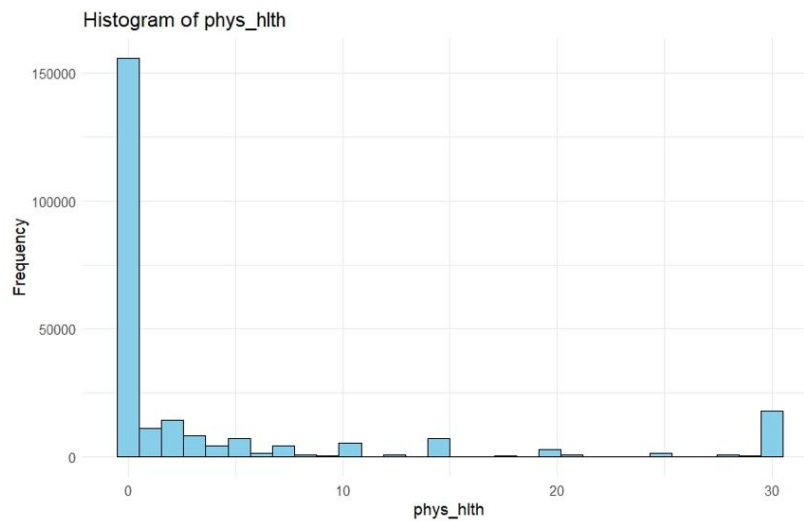
Interpretation: The imbalance justifies the use of oversampling techniques like ROSE to ensure model robustness and mitigate bias toward the majority class.



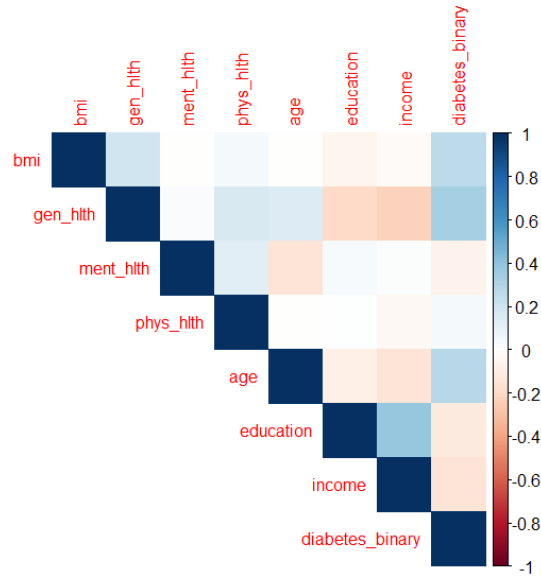
Histogram of BMI Values



Histogram of Mental Health Days



Histogram of Physical Health Days



Correlation Heatmap of Numeric Features

Interpretation: Strong correlations exist between BMI and general health status, reinforcing their roles as crucial predictors in diabetes risk modeling.

4.3 Feature Extraction

Before training any models, we performed feature extraction to organize and prepare the raw dataset. We selected a subset of health-related and demographic variables from the BRFSS 2015 dataset, including indicators such as high blood pressure, high cholesterol, smoking status, physical activity, and healthcare access.

We recast binary features as categorical variables (factors) to properly represent their discrete nature during modeling. We also kept continuous variables such as BMI, mental health days, physical health days, age, education, and income in their numeric form, maintaining their full range for further transformation if needed. No new features were created during extraction; our goal was to cleanly separate categorical and continuous predictors for different modeling requirements.

5. Model Training

5.1 Feature Engineering

During model training preparation, we performed additional feature engineering steps to optimize the dataset for machine learning models. We collapsed the three-class diabetes outcome (*diabetes_012*) into a binary target. This transformation simplified the prediction task and focused the models on detecting diabetic cases.

We also standardized continuous features such as BMI, mental health days, physical health days, age, education, and income. Standardization was necessary to ensure that models like Neural Networks, which are sensitive to feature scale, could converge effectively during training. Finally, we balanced the training set using the ROSE technique.

5.2 Evaluation Metrics

Model performance was assessed using:

- Accuracy - measured the overall proportion of correctly classified cases on the test set.
- Precision - indicated the proportion of predicted diabetic cases that were actually diabetic.
- Recall(Sensitivity) - measured the ability of the model to correctly identify true diabetic cases.
- Area Under the ROC Curve (AUC) - showing how well the model distinguished between diabetic and non-diabetic individuals.
- In addition to overall performance metrics, we also analyzed feature importance to better understand which health indicators contributed most to model decisions. Random Forest feature importance was based on the Mean Decrease in Gini index.

5.3 Model Selection

The following models were trained and evaluated:

- Logistic Regression (baseline linear model)
- Decision Tree (interpretable non-linear model)
- Random Forest (ensemble-based)
- Neural Network (single hidden layer, 5 neurons)

We have selected a variety of machine learning models to compare predictive performance for diabetes detection. We chose Logistic Regression as a simple, interpretable baseline model to understand linear relationships between predictors and the target variable. Decision Trees were selected for their ability to model non-linear interactions and provide easy-to-interpret decision rules.

Random Forests were included to leverage ensemble learning, combining multiple trees to improve robustness and reduce overfitting. Finally, we developed a Neural Network using the Keras framework. To optimize its performance, we fine-tuned the Neural Network across multiple stages, including increasing the number of training epochs, adjusting the learning rate, deepening the architecture, and applying class weighting strategies. These adjustments helped improve recall and overall discrimination performance, making the Neural Network a strong candidate for final model selection.

By selecting models with varying complexity and interpretability, we aimed to systematically compare traditional statistical methods with more advanced machine learning techniques to identify the best-performing approach for early diabetes prediction.

6. Model Validation

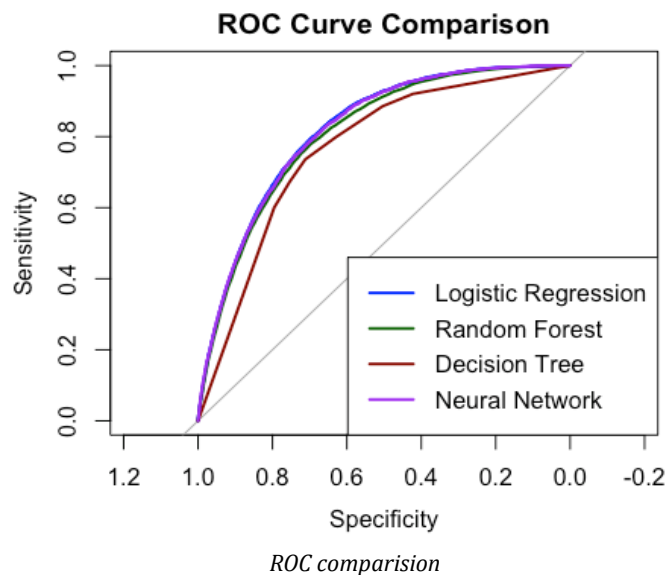
6.1 Testing Results

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	73%	30%	76.5%	0.82
Decision Tree	71.5%	27.8%	73.6%	0.76
Random Forest	78.7%	33.3%	62.5%	0.81
Neural Networks	79.3%	34.1%	61.9%	0.817

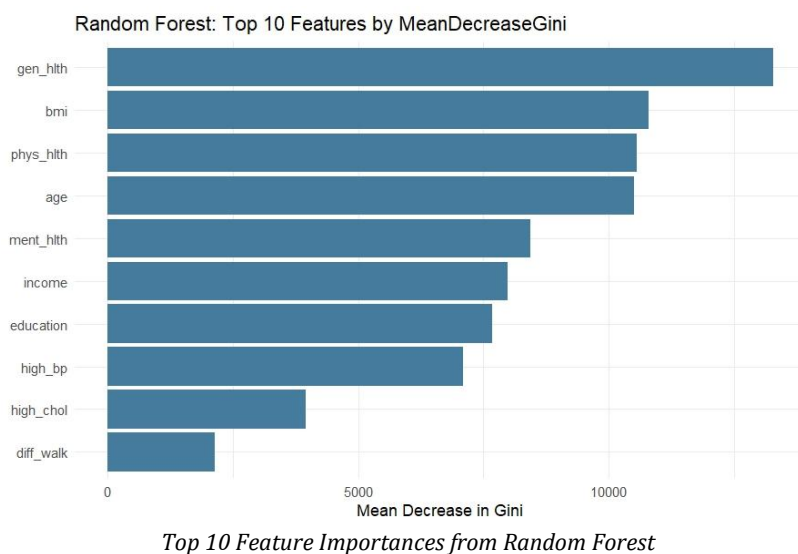
Starting with Logistic Regression, we observed strong baseline performance, in terms of recall and overall separation ability. However, precision remained modest, leading us to explore whether alternative models could improve the balance between sensitivity and specificity. The Decision Tree model, while comparable in recall, showed a slight drop in overall accuracy and discrimination power compared to Logistic Regression. Its tendency to overfit and its weaker separation performance motivated us to seek more stable ensemble approaches. Random Forest improved overall accuracy and precision compared to both Logistic Regression and Decision Tree, but this came at the cost of reduced recall. Given that identifying diabetic cases was a critical objective, this drop in sensitivity prompted further refinement rather than model selection at this stage.

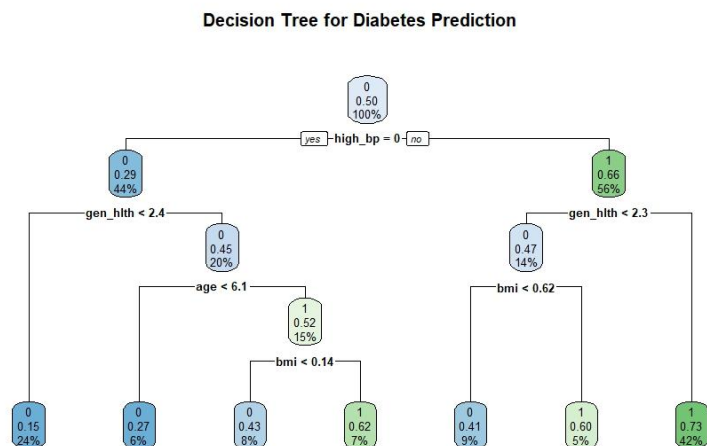
We then moved to developing and tuning a Neural Network model. The initial baseline Neural Network trained for 50 epochs achieved an accuracy of 78.3%, a recall of 59.6%, and an AUC of 0.806. By extending training to 75 epochs, we observed improved AUC (0.817) and slightly better overall accuracy. Threshold lowering and class weighting strategies further boosted recall to 88.4% but significantly decreased precision and accuracy. To achieve the best balance, we implemented a deeper architecture with 64 and 32 neuron layers, applied dropout regularization, and trained for 80 epochs with a reduced learning rate. The final Neural Network model achieved an accuracy of 79.3%, a recall of 61.9%, a precision of 34.1%, and an AUC of 0.81, providing the best trade-off between sensitivity and reliability for diabetes prediction.

ROC curve comparisons supported these results: Neural Network and Logistic Regression curves were closest to the top-left corner, indicating stronger classification performance compared to Random Forest and Decision Tree models.



Feature Importance: The Random Forest model identified several features as highly influential in predicting diabetes. General health status (*gen_hlth*) emerged as the most important variable, followed by body mass index (*bmi*), physical health days (*phys_hlth*), and age (*age*). Mental health days (*ment_hlth*), income, and education also showed substantial contributions, suggesting that broader socioeconomic and health factors play a significant role. Clinical indicators such as high blood pressure (*high_bp*) and high cholesterol (*high_chol*) were also important, along with difficulty walking (*diff_walk*), which may reflect physical limitations related to overall health status. These results align with known risk factors for diabetes and reinforce the importance of both clinical and lifestyle variables in predicting disease risk.





Decision Tree Model Visualization

Interpretation: The tree primarily splits first on features such as Age and General Health, reflecting medically plausible pathways to diabetes risk. However, deeper branches reveal potential overfitting, which limits generalization—an expected limitation in decision tree models without pruning or ensemble techniques.

6.2 Performance Criteria

Given the healthcare focus of this project, Recall and AUC were prioritized over overall Accuracy. Detecting diabetic cases (true positives) was considered more critical than avoiding false positives.

While the Neural Network had slightly lower Recall than Logistic Regression, it offered a better balance between Recall, Precision, and overall Accuracy. Therefore, the Neural Network was selected as the final model based on its robust and reliable performance across multiple evaluation metrics.

6.3 Biases and Risks

- Even after applying the ROSE technique, the original class imbalance toward non-diabetic cases may still influence model performance.
- Self-reported survey data could introduce inaccuracies due to recall errors or biased responses.
- Oversampling methods like ROSE might create synthetic patterns not fully representative of real-world data.
- Despite using early stopping and dropout, the Neural Network remains at some risk of overfitting to noise in the training data.
- Lowering the decision threshold to boost recall led to reduced accuracy and precision, requiring careful trade-offs during real-world deployment.

7. Conclusion

7.1 Positive Results

The Neural Network model, after fine-tuning, provided the best balance between sensitivity, precision, and AUC. Logistic Regression also demonstrated strong baseline performance, confirming the importance of early feature preparation and class balancing efforts. Overall, the models showed strong potential for early identification of individuals at risk for diabetes.

7.2 Negative Results

Despite improvements, all models exhibited moderate precision, indicating a tendency toward false positives. Additionally, sensitivity remained a challenge in certain models like Random Forest, which prioritized overall accuracy at the cost of recall. These results highlight the difficulty of perfectly balancing sensitivity and specificity in healthcare prediction tasks.

7.3 Recommendations

While the final Neural Network model demonstrated strong performance, there remains room for improvement across all key evaluation metrics. Future work could focus on advanced hyperparameter tuning, exploring deeper or alternative network architectures, and applying more sophisticated feature engineering techniques. Further efforts to refine dataset balancing methods may also help enhance model sensitivity and precision. These improvements would contribute to building a more robust and reliable predictive system for accurately identifying individuals at risk of diabetes.

7.4 Caveats/Cautions

The reliance on self-reported survey data introduces potential biases and inaccuracies. Synthetic balancing using ROSE, while helpful for model training, may not perfectly replicate real-world data distributions. Additionally, even with regularization techniques like early stopping and dropout, complex models like Neural Networks remain at some risk of overfitting. Careful validation and cautious interpretation of results are essential when applying these models to new populations.

8. Data Sources

- ❖ Primary Dataset: [Diabetes Health Indicators Dataset \(BRFSS 2015\)](#), available on Kaggle
- ❖ Reference Dataset: [PIMA Indians Diabetes Database](#) for cross-validation.

9. Source Code

- Language: R (v4.3.1)
- Main Script: [Diabetes_risk_prediction.Rmd](#)
- Libraries: tidyverse, caret, randomForest, ROSE, keras, rpart, corrplot, pROC

10. Bibliography

- [1] American Diabetes Association. 2023. "Standards of Medical Care in Diabetes-2023." *Diabetes Care* 46, suppl. 1: S1–S291. <https://doi.org/10.2337/dc23-S001>.
- [2] Ansari, Asif, Manju Nair, and Ramesh Naidu. 2023. "Prediction of Diabetes Mellitus Using Machine Learning Techniques." *Cureus* 15 (2): e34721. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10041290/>.
- [3] Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32. <https://doi.org/10.1023/A:1010933404324>.
- [4] Centers for Disease Control and Prevention. 2025. "Behavioral Risk Factor Surveillance System." Accessed April 22, 2025. <https://www.cdc.gov/brfss/>.
- [5] Kavakiotis, Ioannis, Olga Tsave, Athanasios Salifoglou, Nicos Maglaveras, Ioannis Vlahavas, and Panagiotis Chouvarda. 2017. "Machine Learning and Data Mining Methods in Diabetes Research." *Computational and Structural Biotechnology Journal* 15: 104–16. <https://doi.org/10.1016/j.csbj.2016.12.005>.
- [6] Kuhn, Max, and Kjell Johnson. 2013. *Applied Predictive Modeling*. Springer. Accessed April 15, 2025. https://vuquangnguyen2016.wordpress.com/wp-content/uploads/2018/03/applied-predictive-modeling-max-kuhn-kjell-johnson_1518.pdf.
- [7] Ripley, Brian D. 1996. *Pattern Recognition and Neural Networks*. Cambridge: Cambridge University Press. <https://www.cambridge.org/core/books/pattern-recognition-and-neural-networks/4E038249C9BAA06C8F4EE6F044D09C5C/listing>.
- [8] Teboul, Alex. "Diabetes Health Indicators Dataset." Kaggle. Dataset. Accessed April 20, 2025. <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>.
- [9] UCI Machine Learning Repository. "PIMA Indians Diabetes Database." Kaggle. Dataset. Accessed April 19, 2025. <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.
- [10] Veeranjanyulu, N., K. Rajasekhar, and M. Rama Mohan Babu. 2019. "Predictive Model for Diabetes Mellitus Using Machine Learning Techniques." *BMC Medical Informatics and Decision Making* 19: Article 96. <https://doi.org/10.1186/s12911-019-1004-8>.